



NESR Coiled Tubing Technology



PT. NPS Energy Indonesia
Jakarta, 18 September 2026

Prepared for JDS



Formed in 2017
SPAC NASDAQ
Listed NESR

Acquisition and integration
of Regional Players

Largest
Regional Integrated
Oilfield Services

Head Count
6000+
Across 16 countries



Clients
25 + Customers
(NOC/IOC/ Independents)

2020
ESG Impact I Launched
2023
Rebranded as NEDA

Services Integrated
Production, Drilling
& Evaluation

Technology
Strategic worldwide
partnerships



2016

NPS Energy is established in Indonesia known as **PT. NPS Energy Indonesia**.



2018

- NPS Energy & all the entities globally merged to become NESR.
- 1st Coiled Tubing operations
- 1st Geothermal Hydro Fracturing



2021-2024

- LIU For Capillary Installation & Chemical Injection - Liquid Loaded Wells, Sanga-Sanga
- Geothermal Well Intervention Project: Acidizing , Flow Test & well Logging
- The 1st CO2 Huff and Puff in Pertamina Jatibarang
- EOR & CCUS Pertamina in Sukowati Field



2025-Present

- 1st Offshore Operation - LIU For N2 Unload & Stimulation, Bawean

COMPANY HISTORY

National Energy Services Reunited Corp (NESR) is the largest regional integrated oilfield services provider in MENA, formed in 2017 and listed on NASDAQ. The company employs over 6,000 people across 16 countries.

NESR provides **integrated services** drilling, evaluation, production, and intervention. Its portfolio includes cementing, coiled tubing, fracturing, wireline, well testing, rigs, artificial lift, and pipeline services.



2017

- Obtained Operations Licenses for OFS & Geothermal (GT)
- Acquired **PT Tiger Energy Services ROI**
- 1st Operation in Wireline Geothermal



2018

PT. DFI Asia Energi specializing in stimulation services is acquired.



2019

- 1st Cementing operations
- Sand Jet Perforation
- 1st Multicycle Shear Dilation Fracturing (MCSF) in Langgak Field



2024

- Multicycle Shear Dilation Fracturing (MCSF) in Sanga-Sanga Field



Drill



Drilling and Workover Rigs

Oil and gas well drilling & work overs



Drilling Services and Rentals

Drilling tools & Services



Rig Services

Machine shop & Rig services



Drilling Fluids

Drilling fluid systems

Reach



Directional Drilling

Directional drilling services & solutions



Fishing and Remedials

Fishing, milling, workover, casing repairs



Turbines Drilling

Turbines designed for all drilling conditions

Evaluate and Intervene



Slicklines Services

All slickline applications



Wireline Logging

Open hole & cased hole logging



Well Testing Services

Testing equipment and services

Construct



Cementing

Primary & Remedial Cementing



Laboratory Services

Engineering design & all mobile labs



Completions

Safety systems, HP packers, flow controls

Intervene



Coiled Tubing

Well intervention Services



Stimulation & Pumping

Stimulation Pumping and Frac Services



Nitrogen Services

N2 Equipment and Applications

Produce



Artificial Lifting Services

Wide range of artificial lift solutions



Filtrations Services

Filtration equipment for Fluids



Pipelines Services

Pipeline services to plants & refineries

Well Intervention

Definition

Any operation performed on a well **after** drilling and completion to restore, improve, maintain, or evaluate its production.

Why It's Needed

- Production decline
- Scale, asphaltene, or sand fill
- Formation damage
- Downhole equipment failure
- Water breakthrough or zone isolation

Light Intervention (live well)

Slickline / wireline

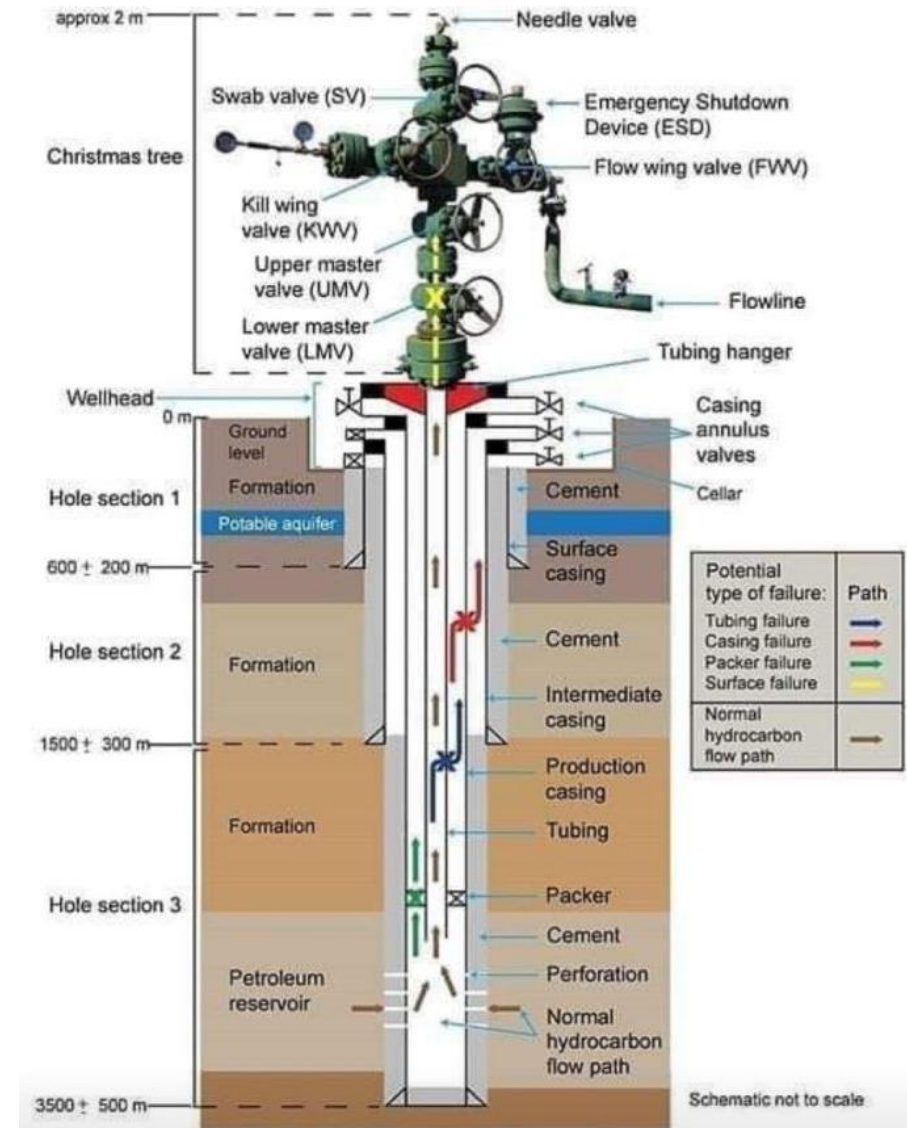
Coiled tubing

Surface pumping & stimulation

Heavy Intervention

Snubbing

Rig workover

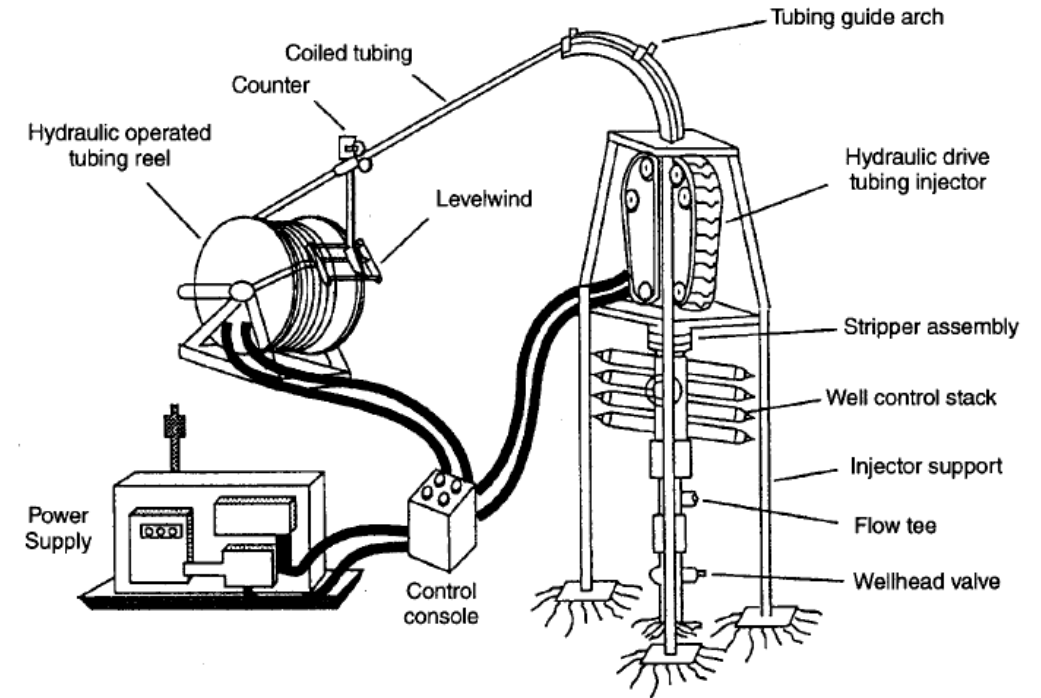


Coiled Tubing

Coiled tubing (CT) is a continuous, jointless steel pipe stored wound on a large reel, which can be run into and out of a well without making up or breaking out connections.

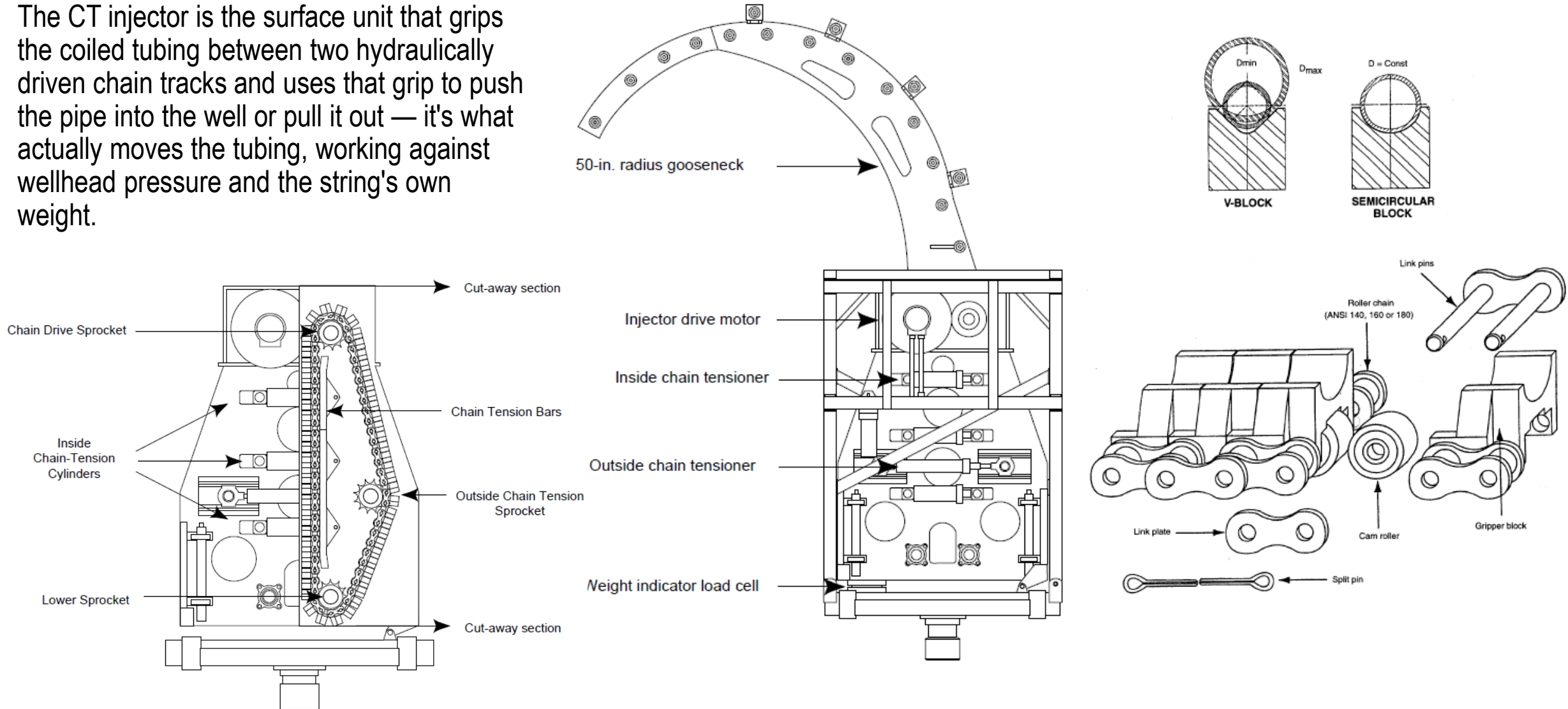
Main surface equipment

- **Injector head** — Grips the pipe with chains and pushes it in or pulls it out
- **Reel** — Stores and spools the tubing
- **Power pack** — Supply hydraulic power
- **Control cabin** — Control the equipment
- **Pressure control equipment** — stripper/packoff and BOP stack, so the well stays live



Coiled Tubing – Injector Head

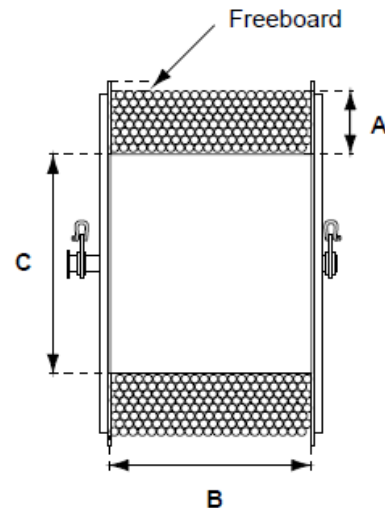
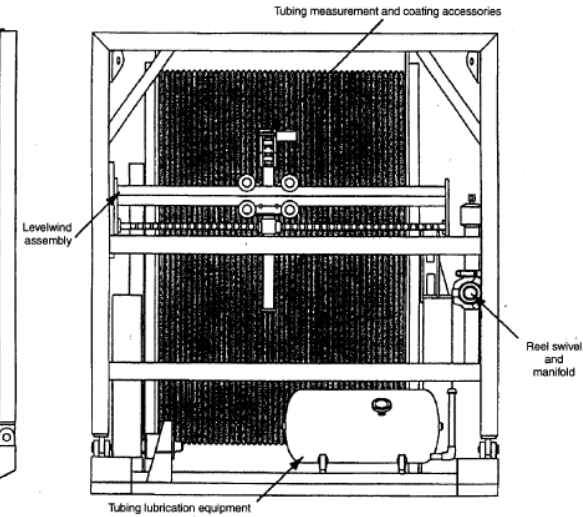
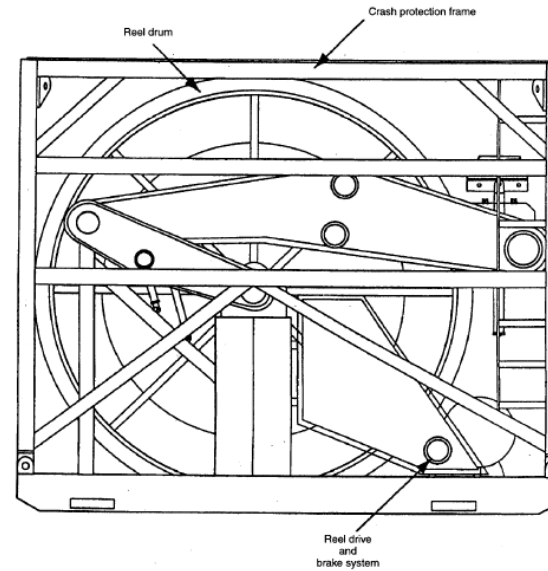
The CT injector is the surface unit that grips the coiled tubing between two hydraulically driven chain tracks and uses that grip to push the pipe into the well or pull it out — it's what actually moves the tubing, working against wellhead pressure and the string's own weight.



Coiled Tubing – Coiled Tubing Reel

The CT reel is used to store and transport the tubing. All of the effort required to run, and retrieve the CT from the well is provided by the injector head. The tension between the reel and injector is necessary to ensure the smooth feeding of the injector and proper spooling onto the tubing reel. The major components of the CT reel include the following.

- Reel drum
- Reel drive system
- Levelwind assembly
- Reel swivel and manifold
- Crash protection frame
- Lubrication tank
- Tubing measurement accessories



$$L = (A + C) (A) (B) (K),$$

where

- L = tubing capacity (ft),
- A = tubing stack height (in.),
- B = width between flanges (in.),
- C = reel drum core diameter (in.), and
- K = K value for different tubing sizes.

K Values for Different Tubing Sizes

Tubing OD (in.)	K Value
1	0.262
1-1/4	0.168
1-1/2	0.116
1-3/4	0.086
2	0.066
2-3/8	0.046
2-7/8	0.032
3-1/2	0.021

Minimum Recommended Freeboard

Tubing OD (in.)	Freeboard (in.)
1 and 1-1/4	1.5
1-1/2 and 1-3/4	2.0
2 and 2-3/8	3.0
2-7/8 and 3-1/2	4.0

Example of Calculating Reel Capacity

Tubing OD = 1.50 in.; therefore, the freeboard is 2.0 in.

A = 22 in. (24-in. drum rim height minus 2-in. freeboard)

B = 72 in.

C = 72 in.

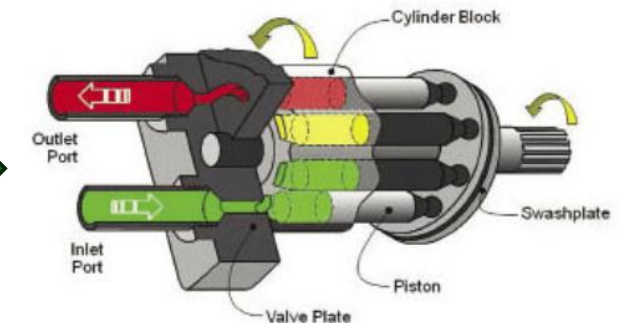
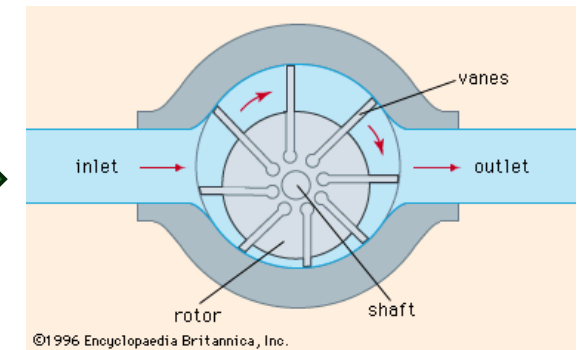
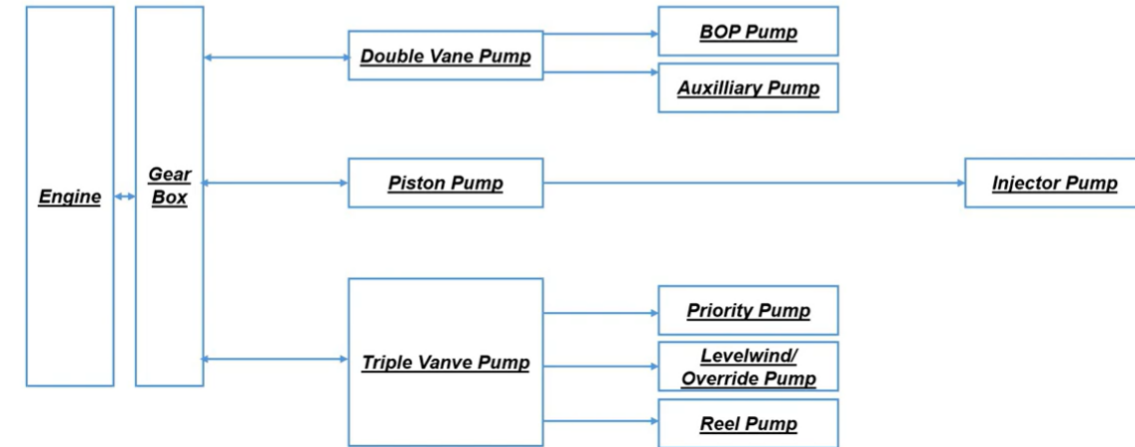
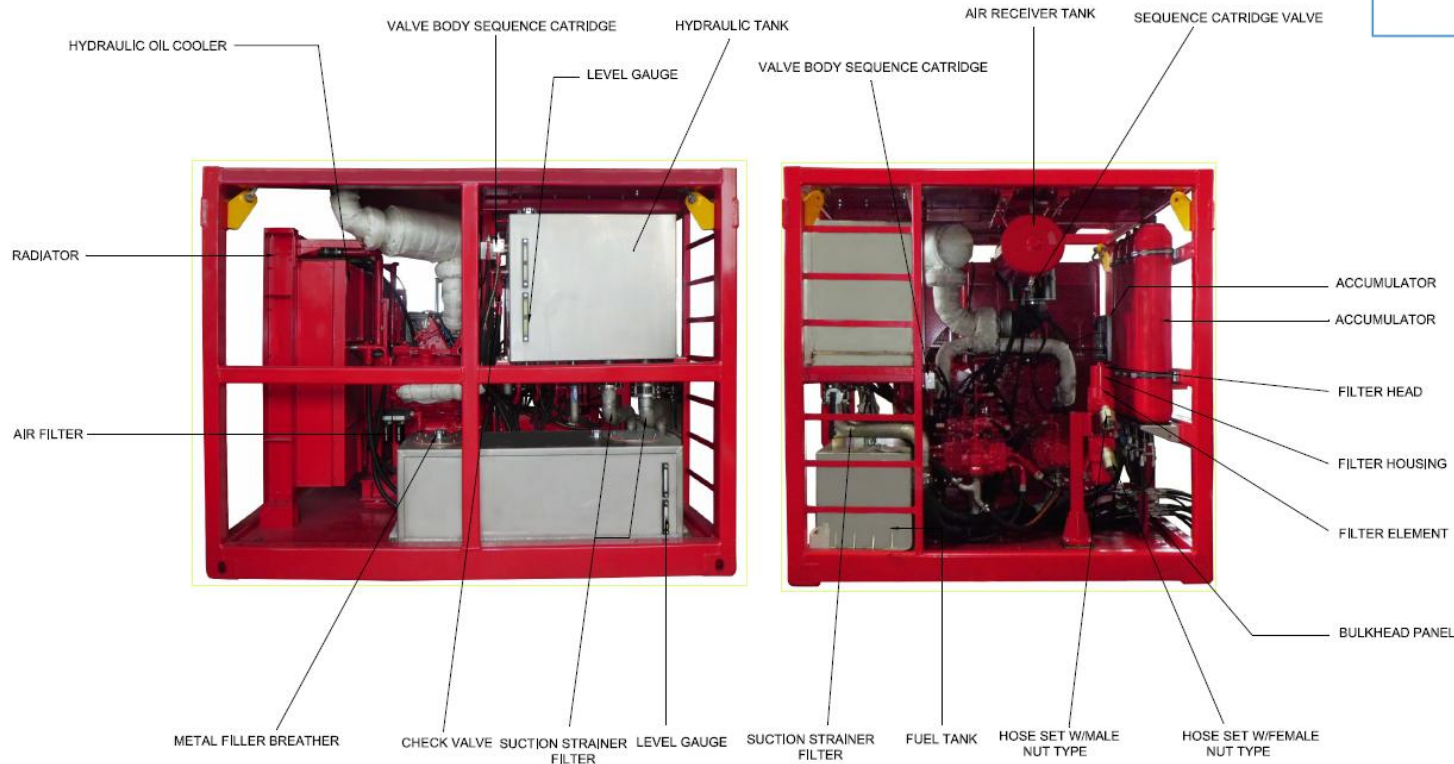
K = 0.116

$$L = (22 + 72) (22) (72) (0.116)$$

$$L = 17,272 \text{ ft of 1.5-in. tubing.}$$

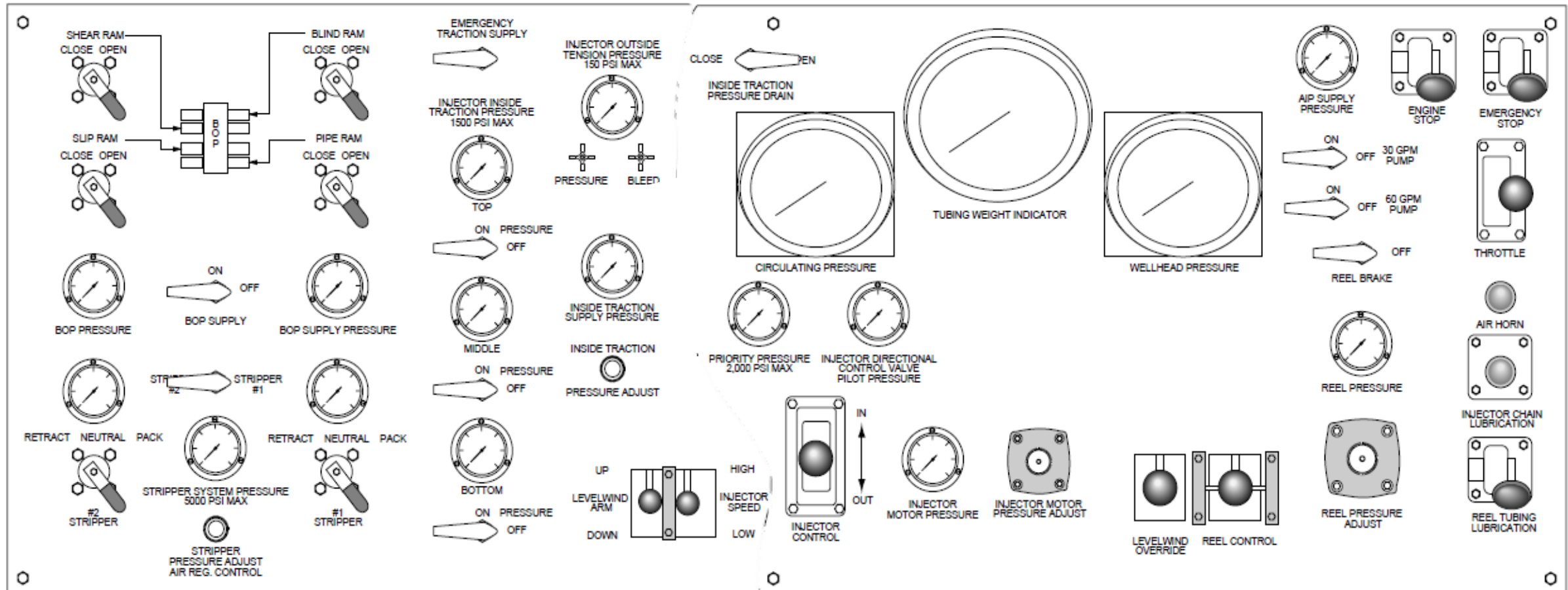
Coiled Tubing – Power Pack

The power pack provides the hydraulic energy to operate the CT unit (CTU) functions and controls. Generally, it consists of a diesel engine driving an array of hydraulic pumps supplying each system or circuit with the required pressure and flow rate.



Coiled Tubing – Control Cabin

The control cabin (control cabin/console) is the operator's station for a CT job — it houses the controls and displays for the injector, reel, BOP stack, and fluid pumping, along with real-time readouts of weight, speed, depth, pressure, and pump rate, so the operator can run the whole job safely from one spot.



Coiled Tubing – PCE

PCE (Pressure Control Equipment)

Equipment stacked on the wellhead to keep the well sealed and under control while CT runs in/out.

Stripper

Dynamic seal — rubber element grips the CT OD while it's moving, sealing pressure as the pipe travels in/out.

BOP (Quad BOP)

Emergency/static barrier, used when stripper can't hold or pipe must be secured/cut. Four rams:

- **Blind** – seals with no pipe in hole
- **Shear** – cuts the pipe
- **Slip** – holds pipe from falling
- **Pipe (CT)** – seals around the pipe, full pressure

Stripper

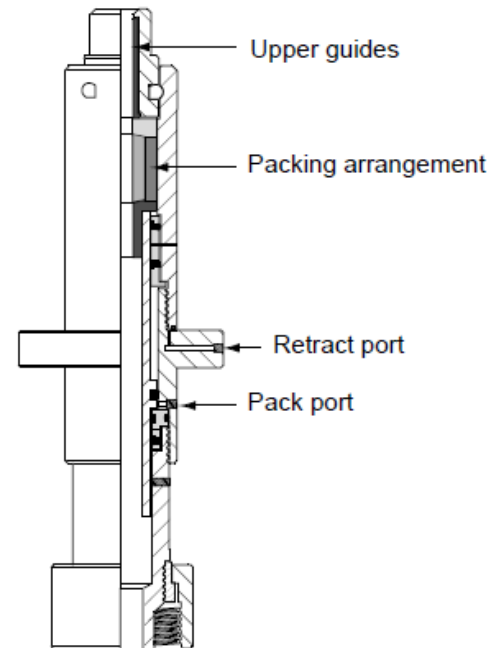


Figure 14. Conventional stripper assembly.

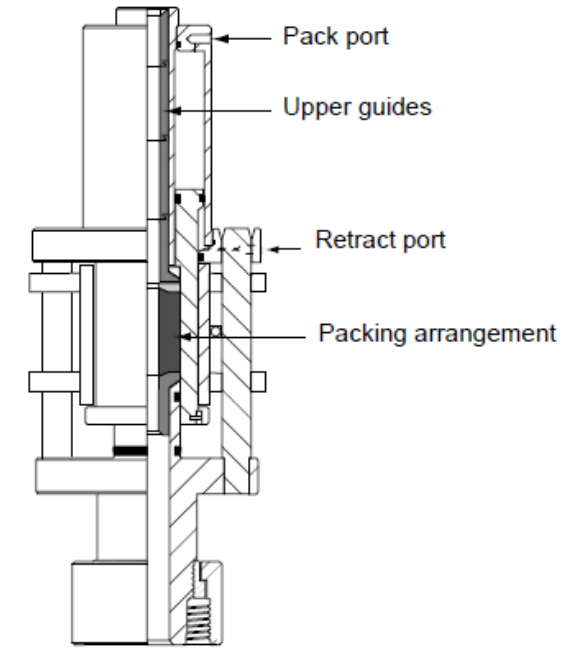


Figure 15. Side-door stripper assembly.

BOP

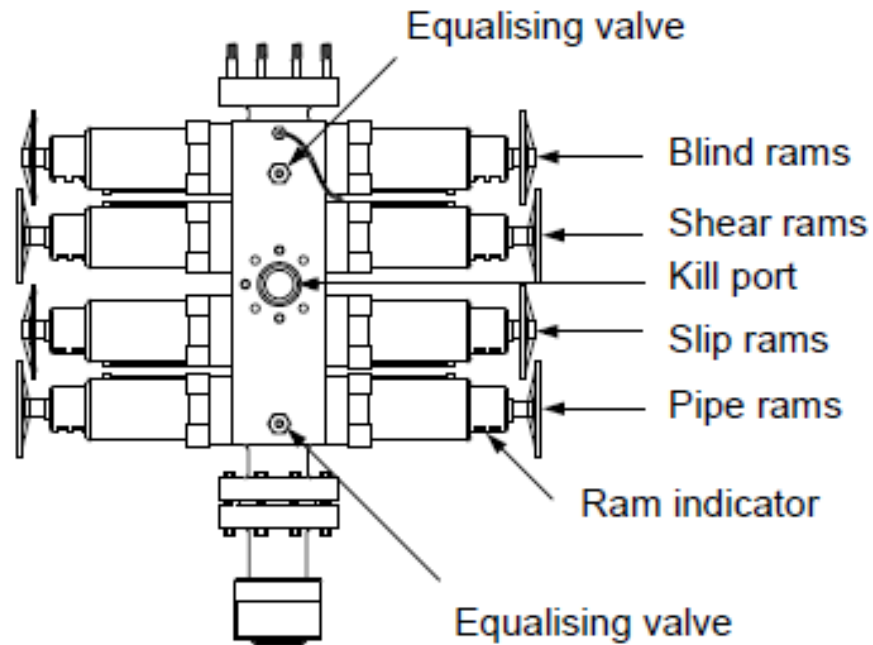


Figure 16. Quad BOP

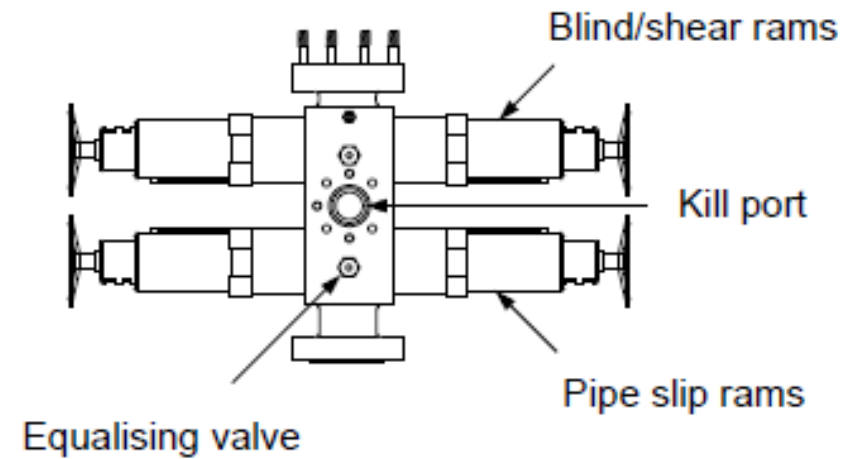


Figure 17. Combi BOP

Quad BOP Rams

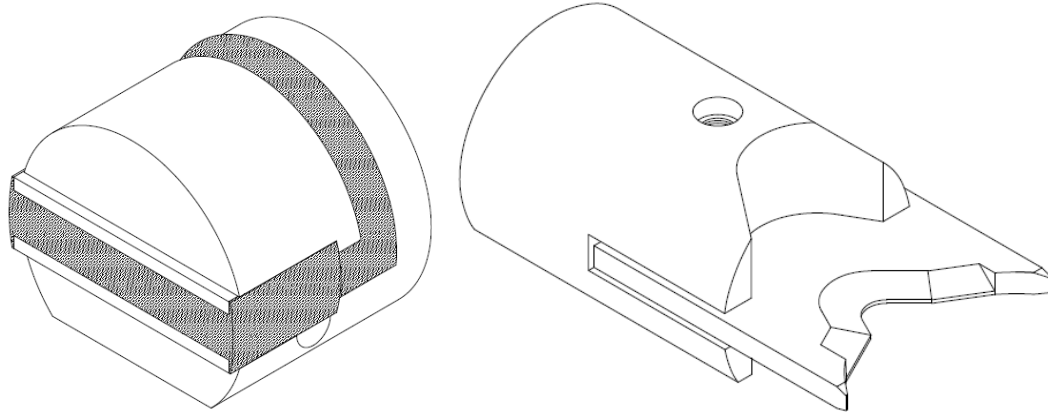


Figure 11—Blind Ram

Figure 13—Shear Rams

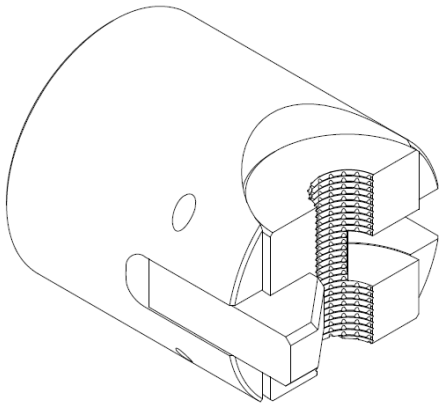


Figure 14—Slip Rams

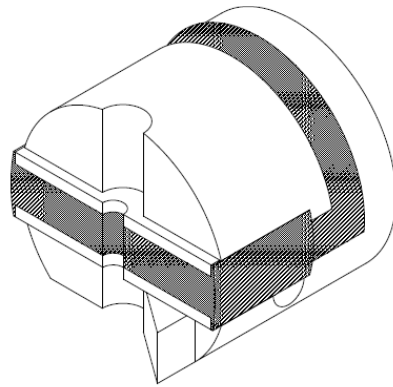


Figure 12—Pipe Rams

Combi BOP Rams

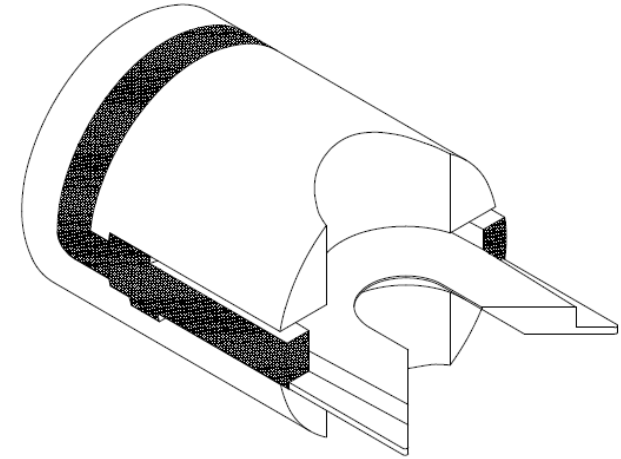


Figure 15—Shear-blind Combination Rams

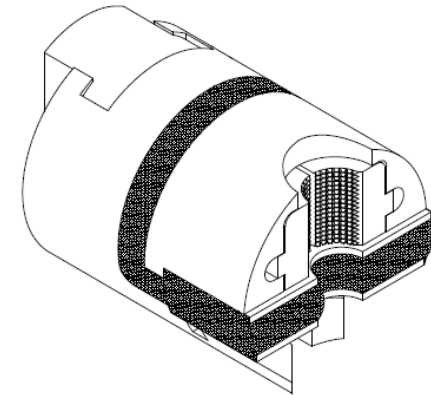


Figure 16—Pipe-slip Combination Rams

Well control equipment shall be identified, installed, tested, and used to promote and maintain control of the well at all times. The following issues should be reviewed to ensure compliance with this requirement:

- a) maximum anticipated surface pressure (MASP);
- b) maximum anticipated operating pressure (MAOP);
- c) well control barriers;
- d) CT operational pressure categories;
- e) well control stack configurations;
- f) bore size, rated working pressure, and connections of well control equipment.

Maximum Anticipated Surface Pressure (MASP)

The MASP is the highest pressure predicted to be encountered at the surface of a well. This pressure prediction should be based upon formation pressure minus a wellbore filled with native formation fluid at current conditions. If formation fluid information is unknown; this pressure prediction should be based upon formation pressure minus a wellbore filled with dry gas from the surface to the completion interval.

Maximum Anticipated Operating Pressure (MAOP)

The MAOP for a given piece of equipment is the highest calculated pressure that a given equipment component will be subjected to during the execution of the prescribed service and/or during a contingency operation.

NOTE: The MAOP may be equal to or greater than the MASP.

Coiled Tubing – Well Control

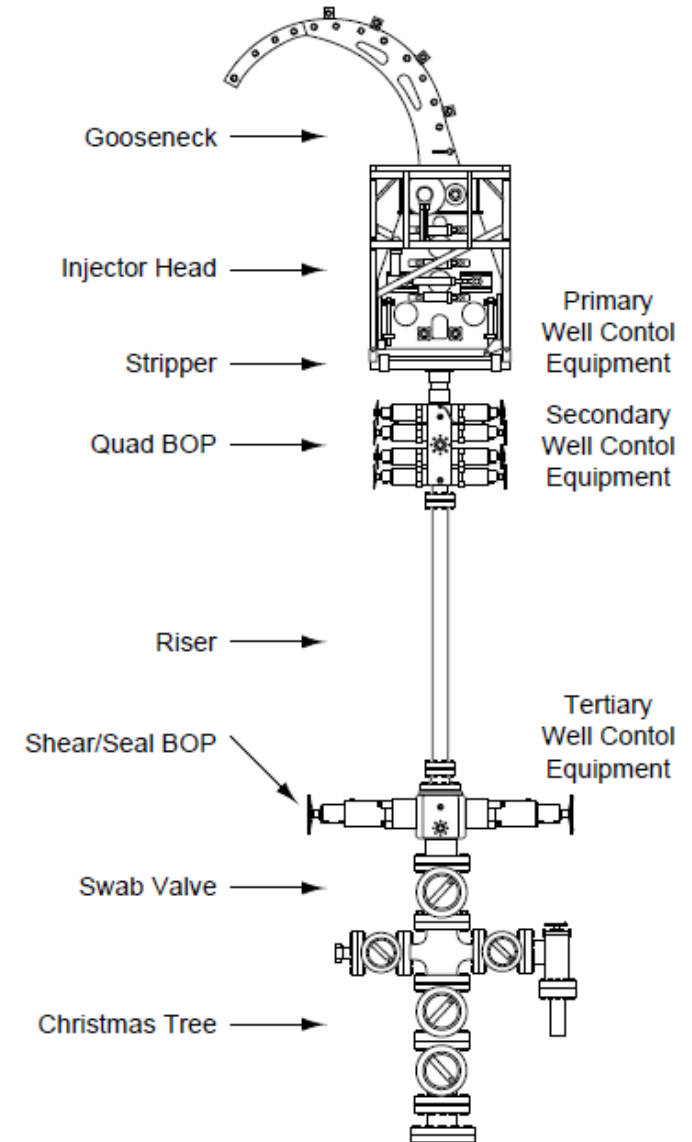
Excluding PC 0, the standard well control stack configuration for all pressure categories shall include the following components. The recommended order of the components (from the top down) is shown as follows:

- 1) one stripper or annular preventer-type well control component
- 2) one blind ram well control component
- 3) one shear ram well control component
- 4) one kill line inlet
- 5) one slip ram well control component
- 6) one pipe ram well control component

Pressure Category (PC)	MASP Range psig	Minimum Rated Working Pressure of Stack psig	Minimum Number of Barriers
PC-0	0 ^a	3000	1
PC-1	1 to 1500	3000	2
PC-2	1501 to 3500	5000	2
PC-3	3501 to 7500	10,000	2
PC-4	7501 to 12,500	15,000	2

The minimum rated working pressure of the stack shall be equal to or greater than the MAOP.

^a PC-0 applies to those wells demonstrated as incapable of unassisted flow to surface (based on local regulatory agency guidelines).



Coiled Tubing – Well Control (PC-0 : 0 Psig)

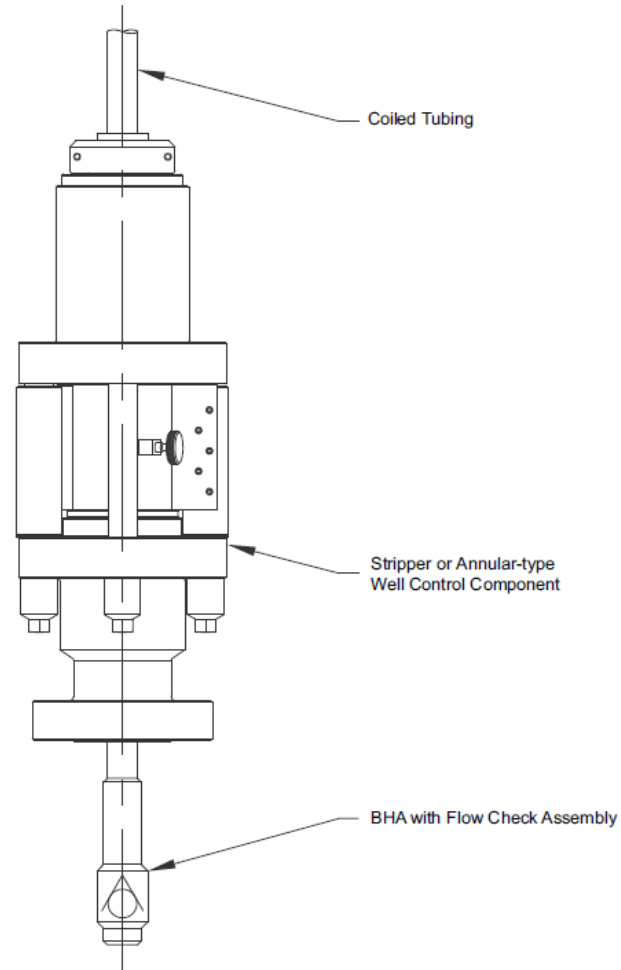


Figure 1—Well Control Stack Components for PC-0 Conditions—for Returns Taken Through the Tree or Wellhead Below the Well Control Stack

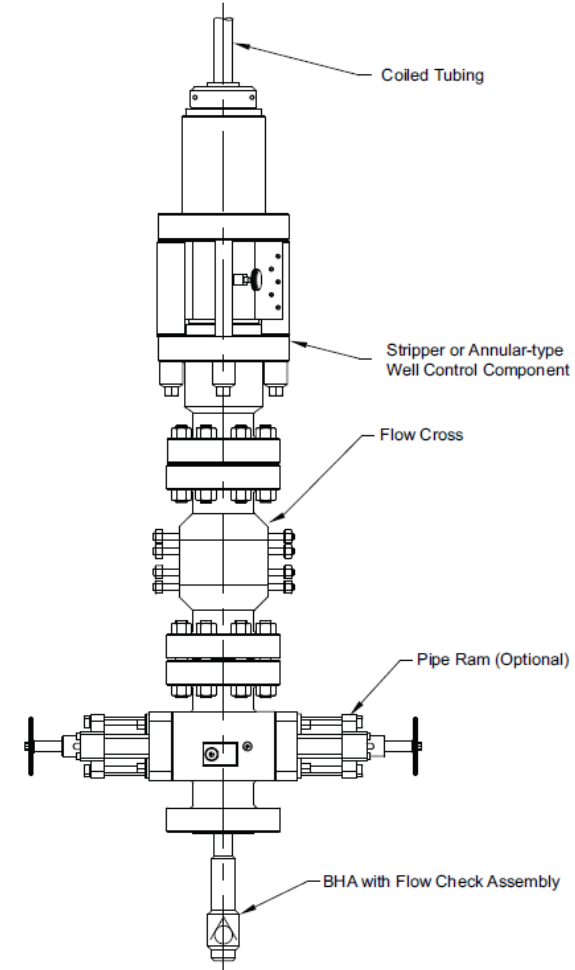


Figure 2—Well Control Stack Components for PC-0 Condition—for Returns Taken Through the Flow Cross or Flow Tee Installed in the Well Control Stack

Coiled Tubing – Well Control (PC-1 : 1 - 1500 Psig)

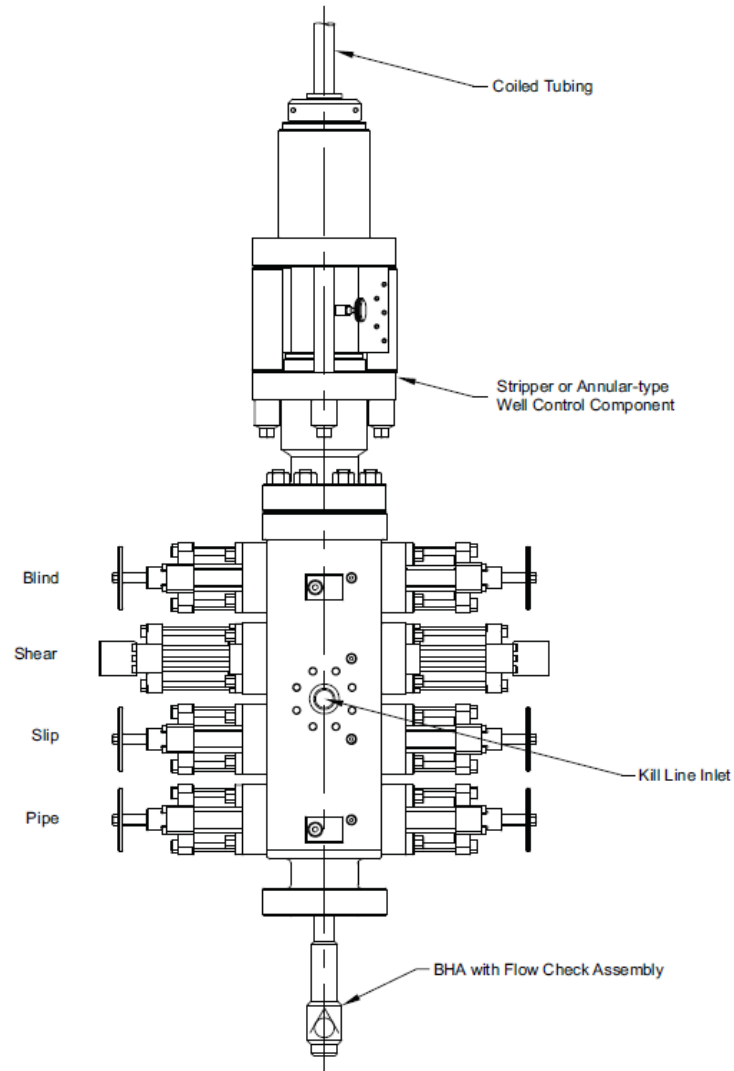


Figure 3—Well Control Stack Components for PC-1 Conditions (1 psig to 1500 psig)—for Returns Taken Through the Tree or Wellhead Below the Well Control Stack

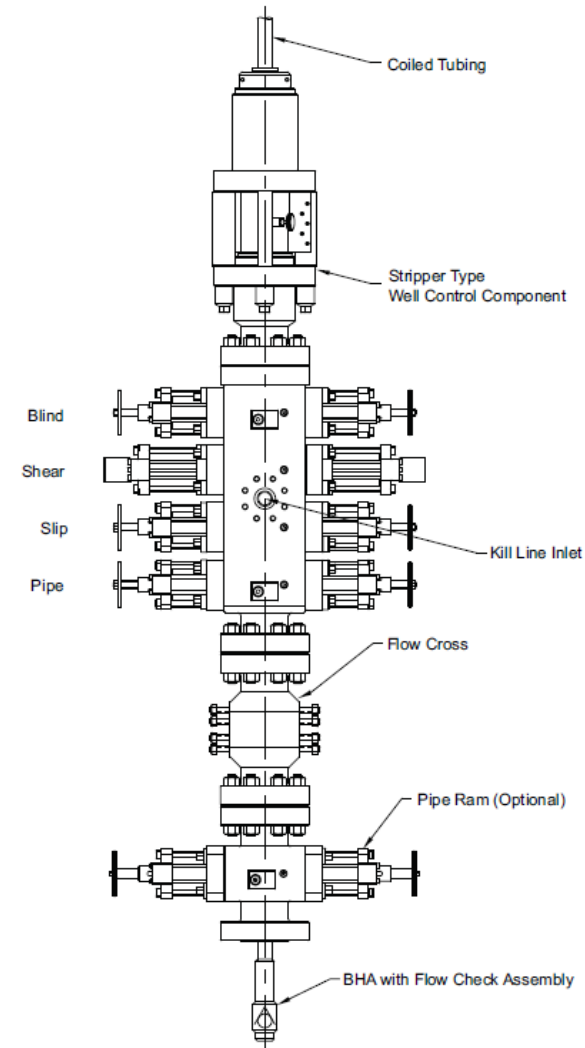


Figure 4—Well Control Stack Components for PC-1 Conditions (1 psig to 1500 psig)—for Returns Taken Through the Flow Cross or Flow Tee Installed in the Well Control Stack

Coiled Tubing – Well Control (PC-2: 1501 - 3500 Psig)

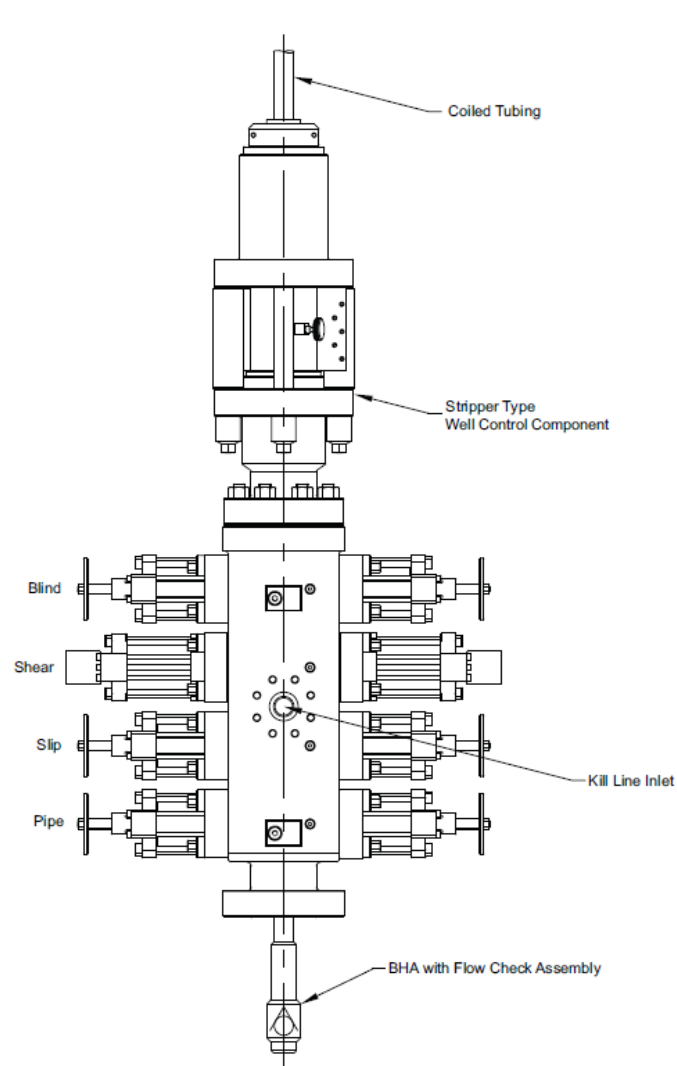


Figure 5—Well Control Stack Components for PC-2 Conditions (1501 psig to 3500 psig)—for Returns Taken Through the Tree or Wellhead Below the Well Control Stack

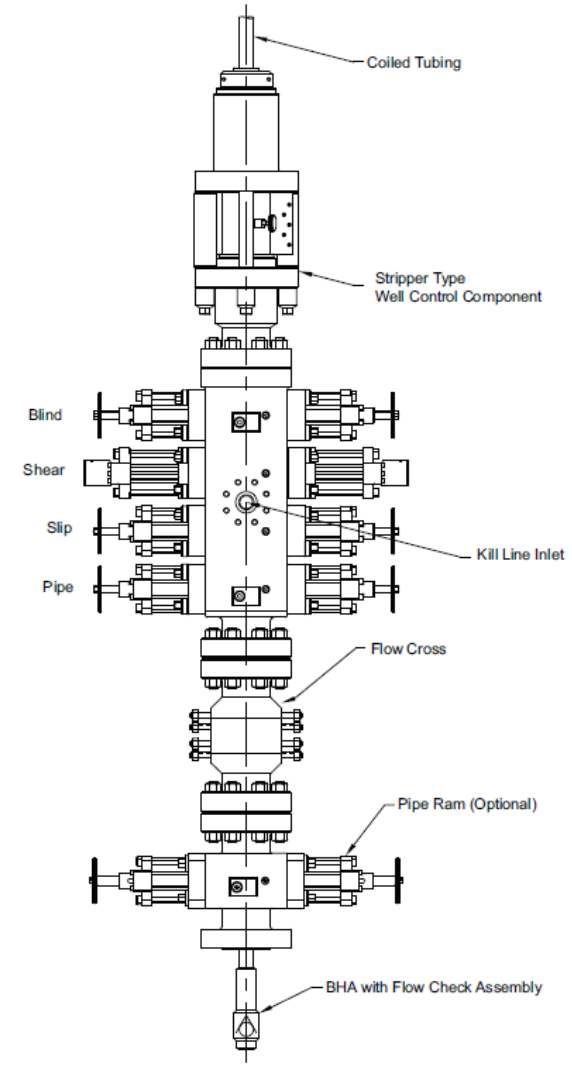


Figure 6—Well Control Stack Components for PC-2 Conditions (1501 psig to 3500 psig)—for Returns Taken Through the Flow Cross or Flow Tee Installed in the Well Control Stack

Coiled Tubing – Well Control (PC-3: 3501 - 7500 Psig)

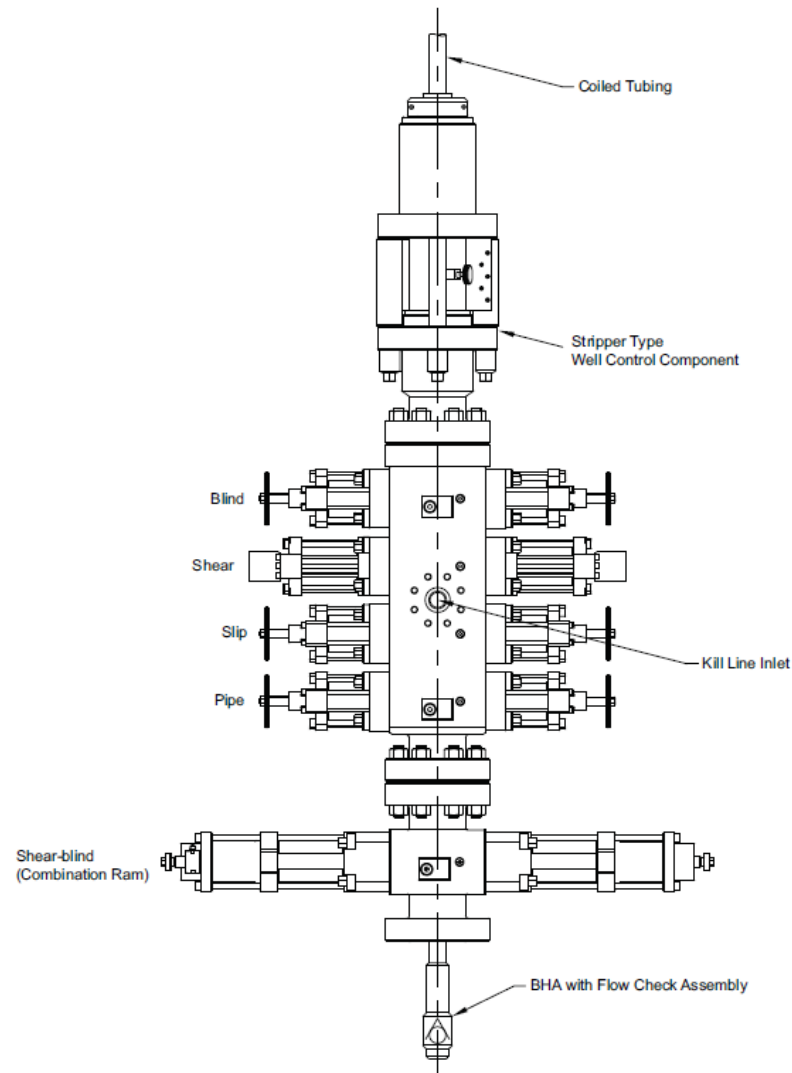


Figure 7—Well Control Stack Components for PC-3 Conditions (3501 psig to 7500 psig)—for Returns Taken Through the Tree or Wellhead Below the Well Control Stack

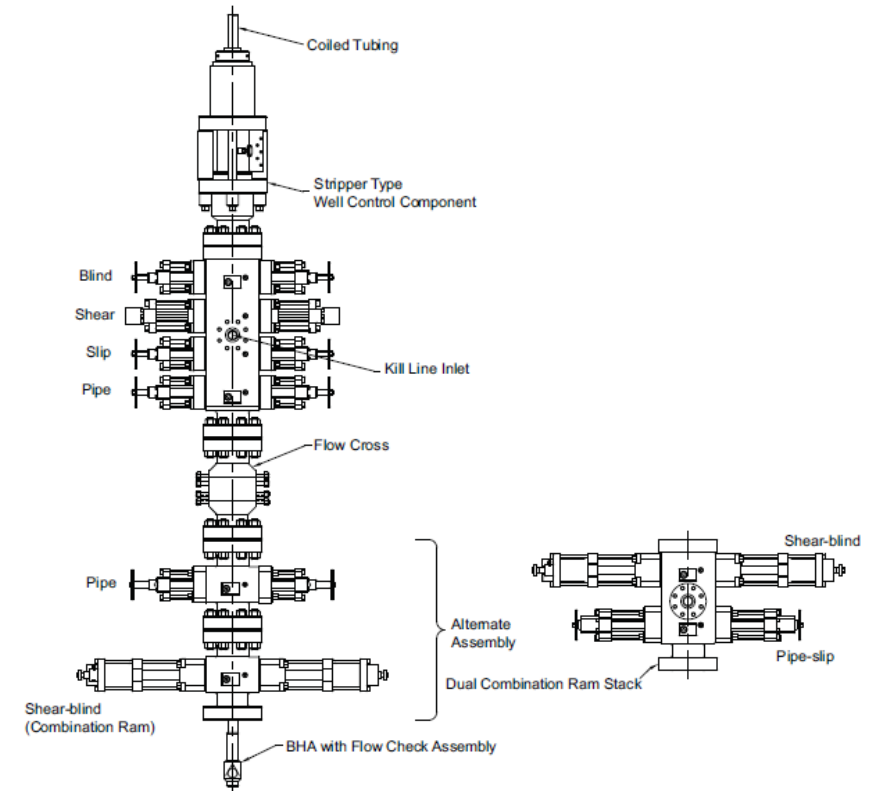


Figure 8—Well Control Stack Components for PC-3 Conditions (3501 psig to 7500 psig)—for Returns Taken Through the Flow Cross or Flow Tee Installed in the Well Control Stack

Coiled Tubing – Well Control (PC-3: 7501 – 12,500 Psig)

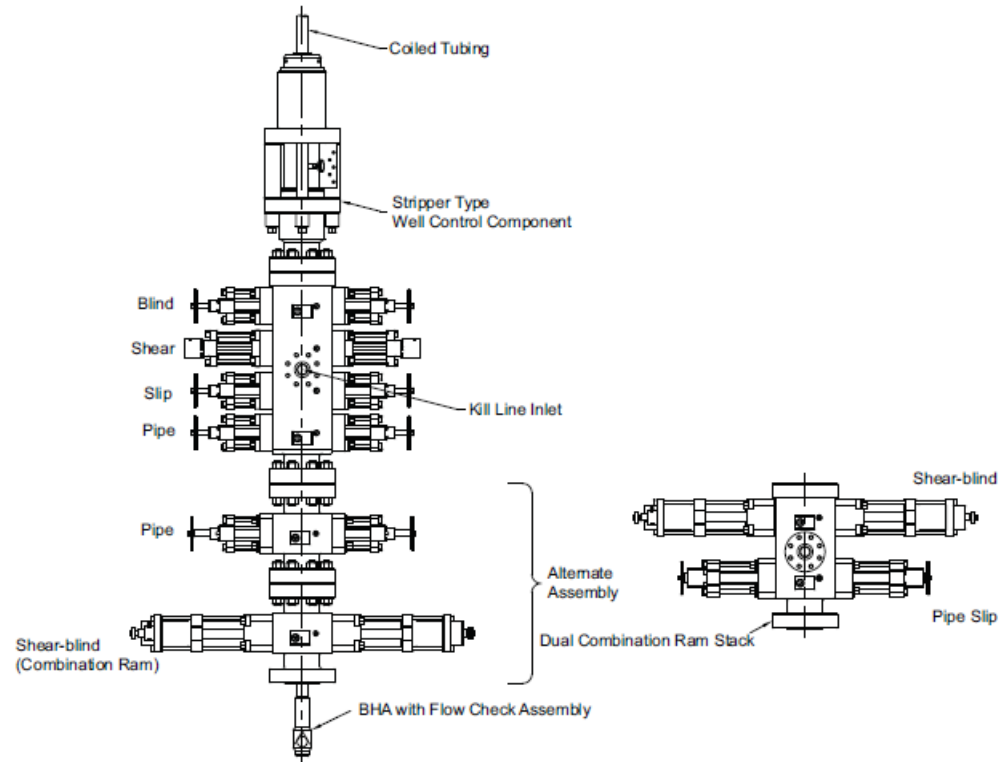


Figure 9—Well Control Stack Components for PC-4 Conditions (7501 psig to 12,500 psig)—for Returns Taken Through the Tree or Wellhead Below the Well Control Stack

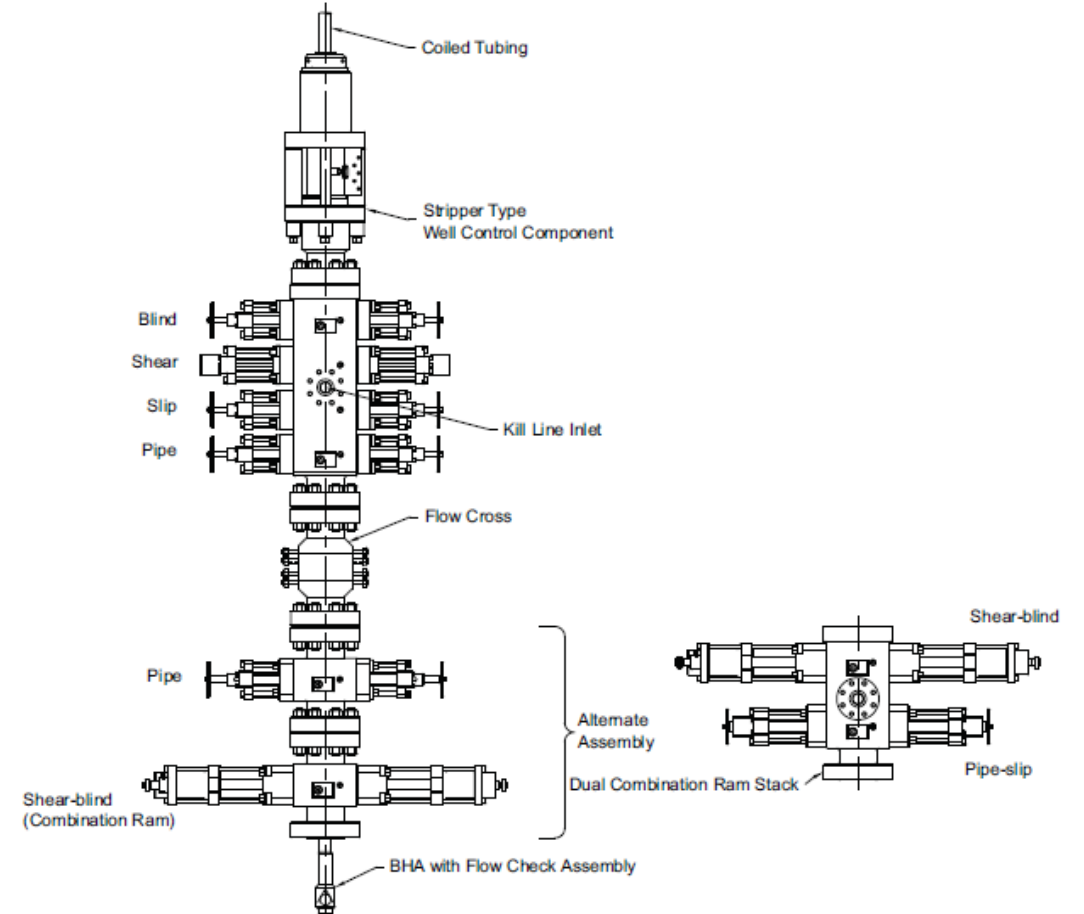


Figure 10—Well Control Stack Components for PC-4 Conditions (7501 psig to 12,500 psig)—for Returns Taken Through the Flow Cross or Flow Tee Installed in the Well Control Stack

COILED TUBING UNIT #1 SPECIFICATION



Control Cabin

- Control Panel for all units
- Manual pressure gauge (WHP, CP, Weight Indicator)
- Equipped with a digital data monitoring and recorder MEDCO

Mechanical Depth Counter

Gooseneck

- 72" radius

Injector Head

- Model: Stelkraft 660
- Maximum pull: 60K lbs
- Gripper block 1.5"

CT Reel

- Model: WRN18A
- Reel Capacity: 15000 ft of 1.5"
- String Model: GT-90
- String Wall Thickness: 0.175" – 2"
- String Yield Strength: 90K psi

Power Pack

- Model: 6V-92T Detroit Diesel Engine
- Turbocharged
- 550 HP @2100 RPM

Blow Out Preventer (BOP)

- Model: 10K Quad BOP
- Size: 3.06"
- Working Pressure: 10K psi



COILED TUBING UNIT #2 SPECIFICATION



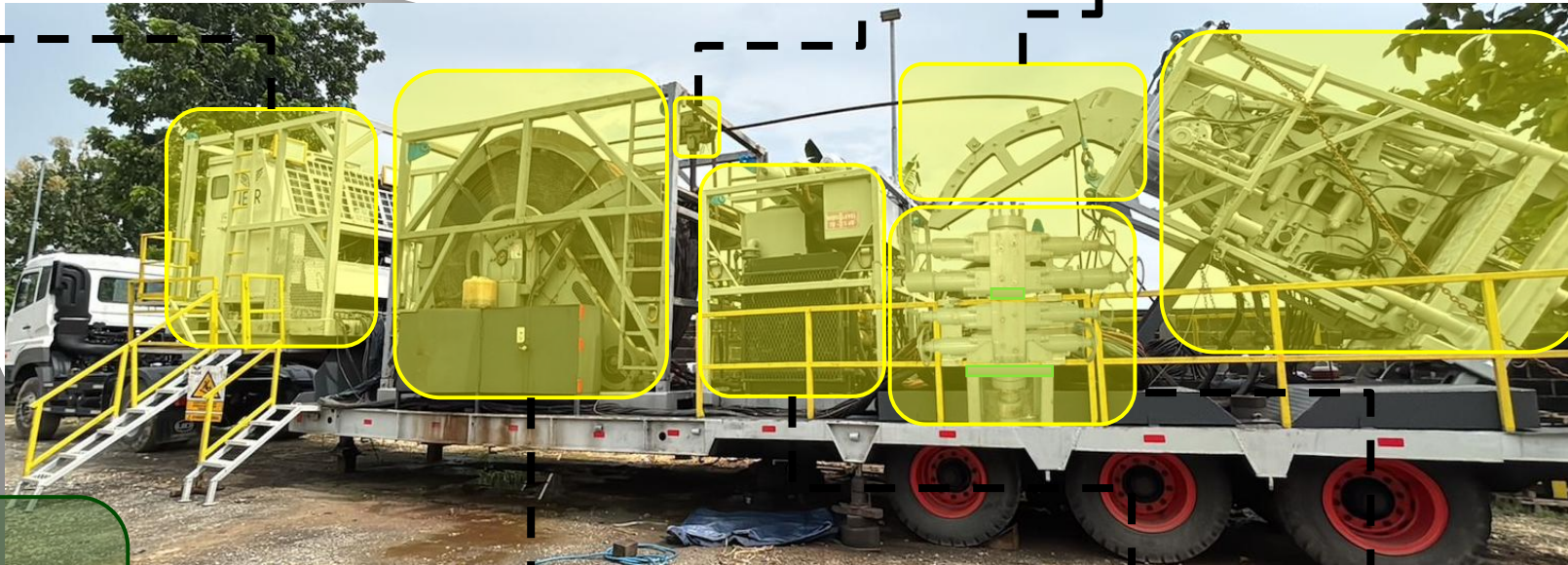
Control Cabin

- SN: CC122
- Data Acquisition (MEDCO)
- Control Panel for all units
- Manual pressure gauge (WHP, CP, Weight Indicator)

Mechanical Depth Counter

Gooseneck

- 72" radius



Injector Head Hydra Rig

- Model: HR-240 (IH-122)
- Maximum pull: 40K lbs
- Open and close the loop hydraulic system

CT Reel

- SN: WR122
- Model: Hydrarig RC-200
- Reel Capacity: 15000 ft of 1.5"
- String Model: QT-800
- String Wall Thickness: 0.145"
- String Yield Strength: 85K psi

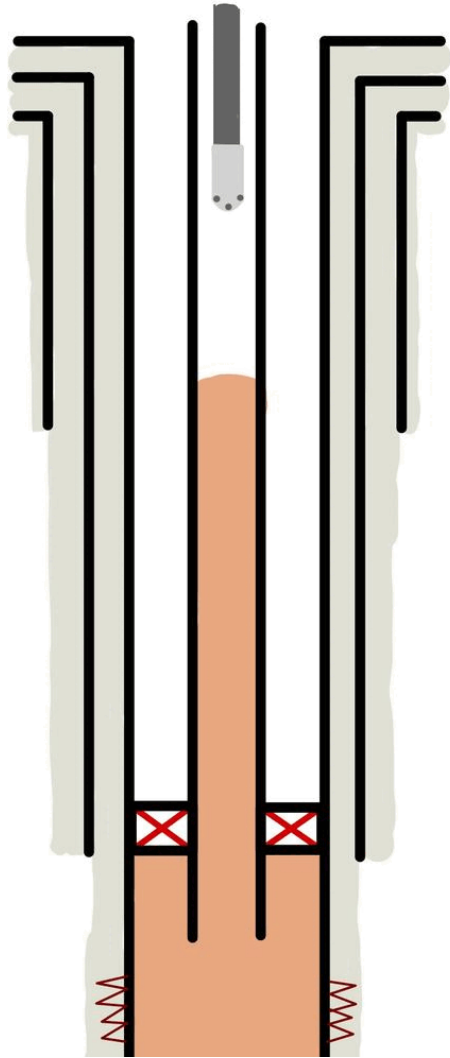
Power Pack

- SN: PP122
- Model: 6V-92T Detroit Diesel Engine
- Turbocharged
- 550 HP @2100 RPM

Blow Out Preventer (BOP)

- SN: BOP-122
- Model: EH44 10K Quad BOP
- Size: 4.06"
- Working Pressure: 10K psi
- Hydraulic Pressure: 3K psi

COILED TUBING OPERATION



Nitrogen Unloading

N₂ Coiled Tubing is used to perform well interventions without shutting down production. The N₂ unloading operation effectively reduces hydrostatic pressure by displacing higher-density fluids, enhancing hydrocarbons' ability to reach the surface.

The primary objectives include:

- Well Cleaning
- Lifting Fluid
- Pressure Reduction

Advantage

- Inert and Non-Corrosive
- Quick and Efficient
- Environmentally Friendly
- Real-Time Monitoring and Adjustments

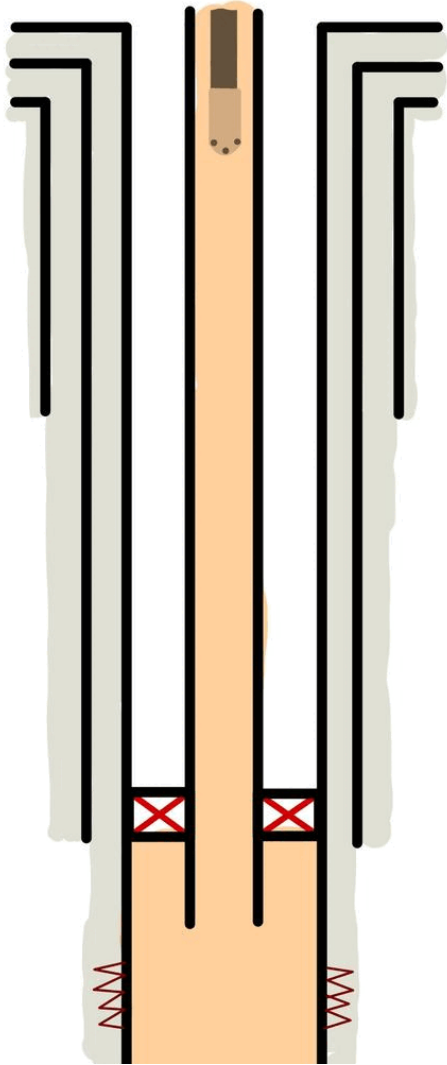
Equipment

- Coiled Tubing Unit
- Nitrogen Converter Unit
- Nitrogen Iso-Tank

Material

- Liquid Nitrogen

COILED TUBING OPERATION



Matrix Acidizing

Matrix acidizing is a well stimulation process used to improve the permeability of the reservoir rock and enhance hydrocarbon flow into the wellbore. It is performed below the formation's fracture pressure, meaning it does not create new fractures but instead cleans or dissolves materials that restrict flow.

Advantage

- Cost Effective
- Shorter Downtime
- Environmentally Safer
- Precise Acid Placement with CTU
- Real-Time Monitoring and Adjustments

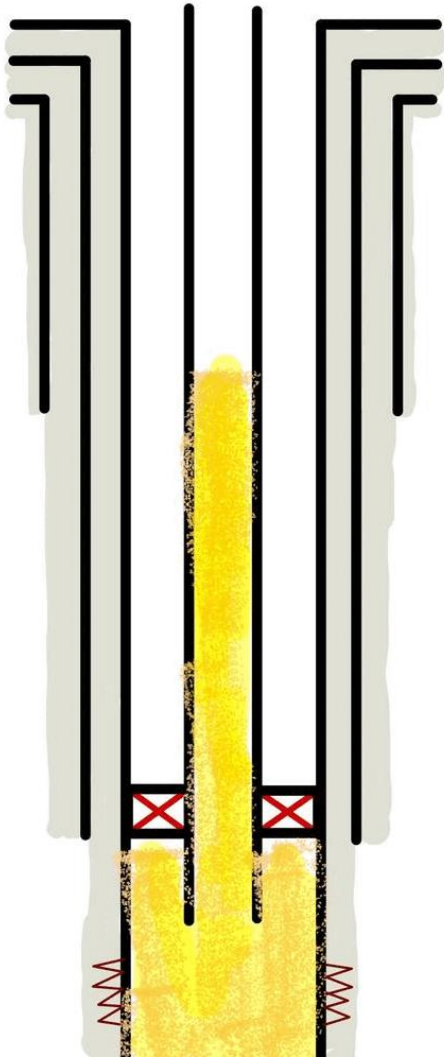
Equipment

- Coiled Tubing Unit
- Triplex Pump Unit
- Acid Tank
- Water Tank

Material

- HCL 32%
- Corrosion Control Additives
- Iron Control Additives
- Diverter Agent

COILED TUBING OPERATION



Sand Clean Out

Sand cleanout is a well intervention technique designed to remove sand and solid debris that accumulate within the wellbore due to hydraulic fracturing or other cause, which can obstruct fluid flow and reduce production efficiency.

Advantage

- Precise Depth Control
- Continuous Circulation Capability
- Reduced Operational Time and Cost
- Real-Time Monitoring and Adjustments

Equipment

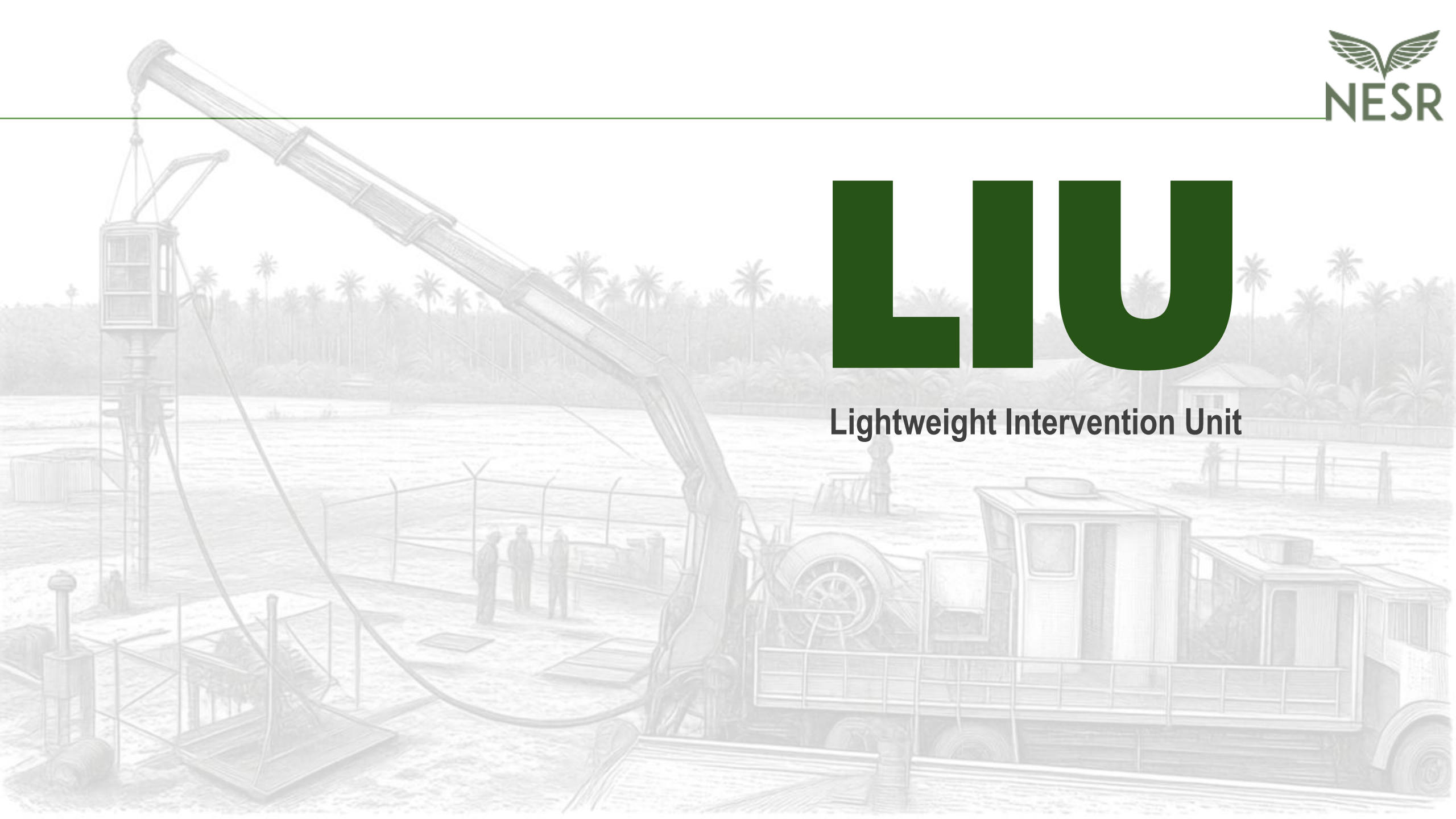
- Coiled Tubing Unit
- Triplex Pump Unit
- Chemical Tank
- Water Tank

Material

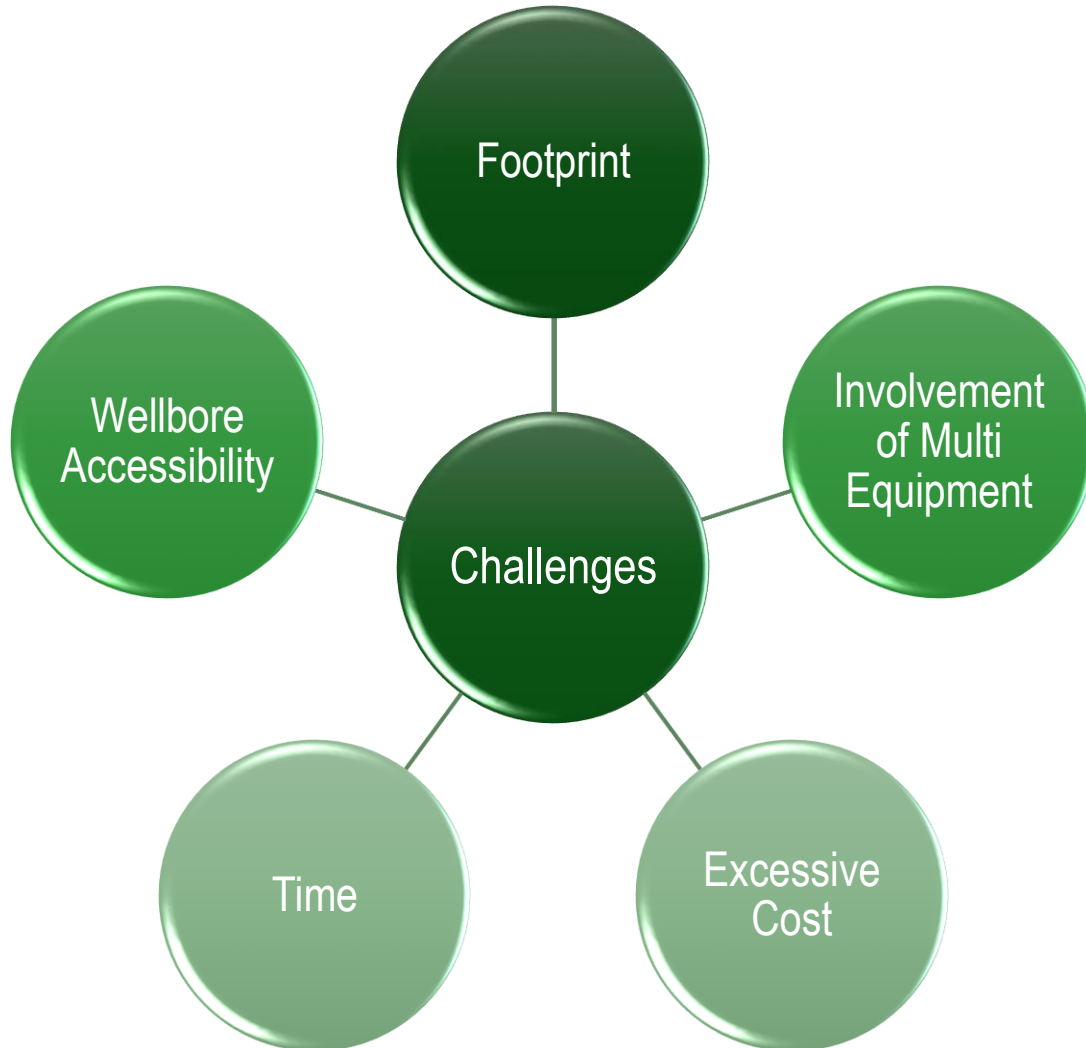
- Gelling Agent
- Corrosion Control Additives
- Surfactant
- Foaming Agent (for N₂ Foam Cleanouts)

LIU

Lightweight Intervention Unit



Well Intervention Challenges



- Mobilization , Setup & Rig up is time consuming
- Involving many equipments in different package lead to high cost , space and ineffective execution time
- Limitation on wellbore condition makes rig operation become mandatory

LIU – To Meet Well Intervention Needs

7 X 2.5 X 3.2 M (LXWXH)

Modular
(to meet limited well-site
dimention)



Footprint

Involvement of Multi Equipment

Built in Engine,
Power Pack,
Cabin, DAS, Reel,
Tank, Pump,
Injector, PCE



More effective in
Mobilization Cost,
Total Service Charge
and Operation Days Value

Excessive Cost

Time

Reduce Mobs , Setup and
Rig up Time



Wellbore Accessibility

Smaller ID and lighter for versatile operation,
Replaceable Size &
Type of intervention
5-7 ton



LIU – Main Features



LIU – Main Features

3 10 Ton Crane



7 Gooseneck

8 Injector applicable
For 0.25-0.75"

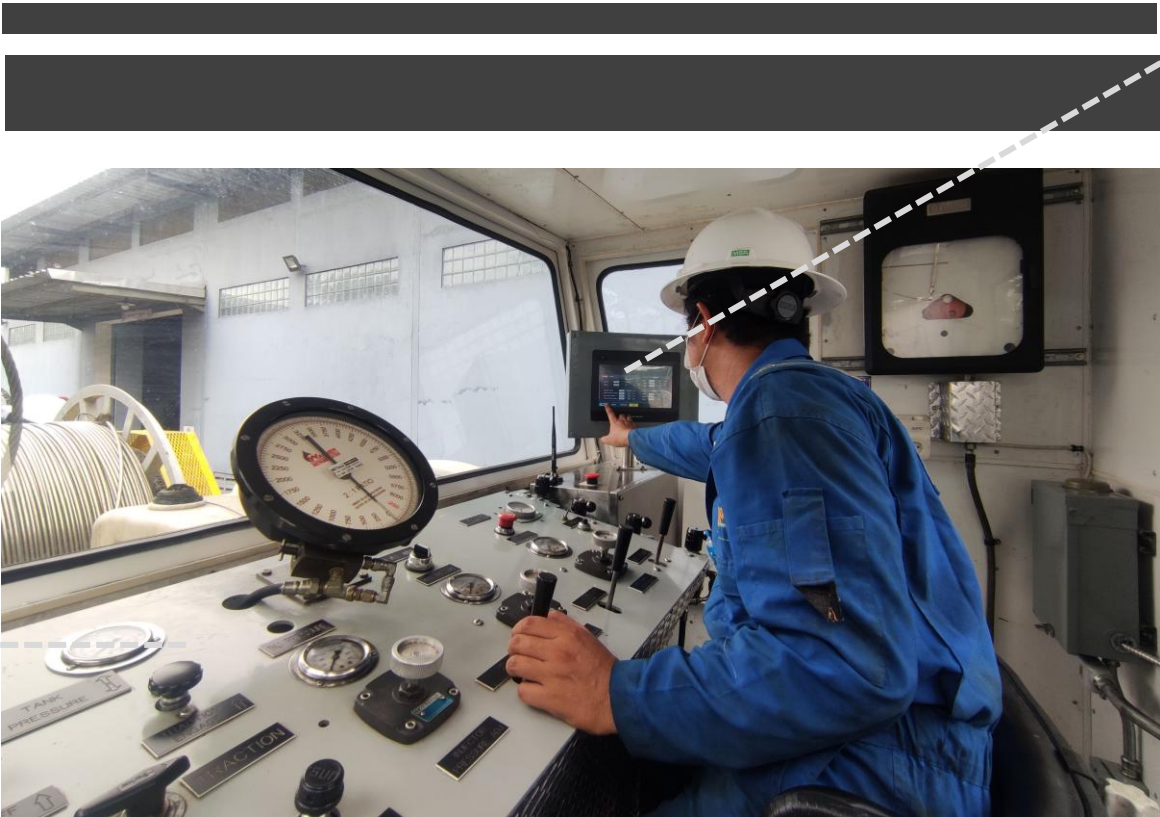
9 Electrical Genset

4 Replaceable &
Detachable Reel

LIU – Main Features



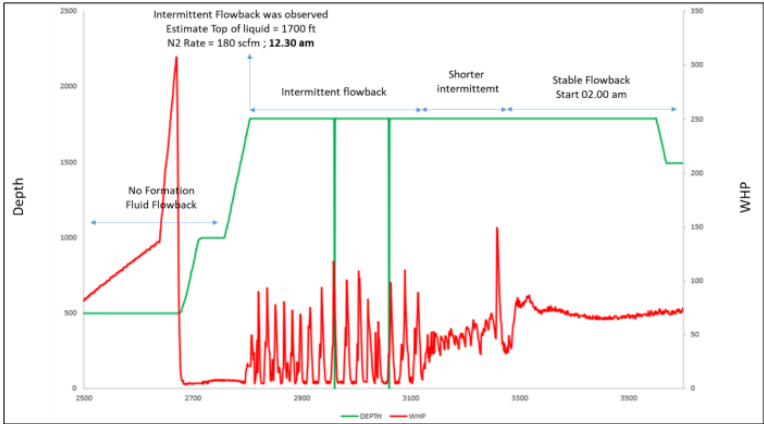
Inside The Cabin



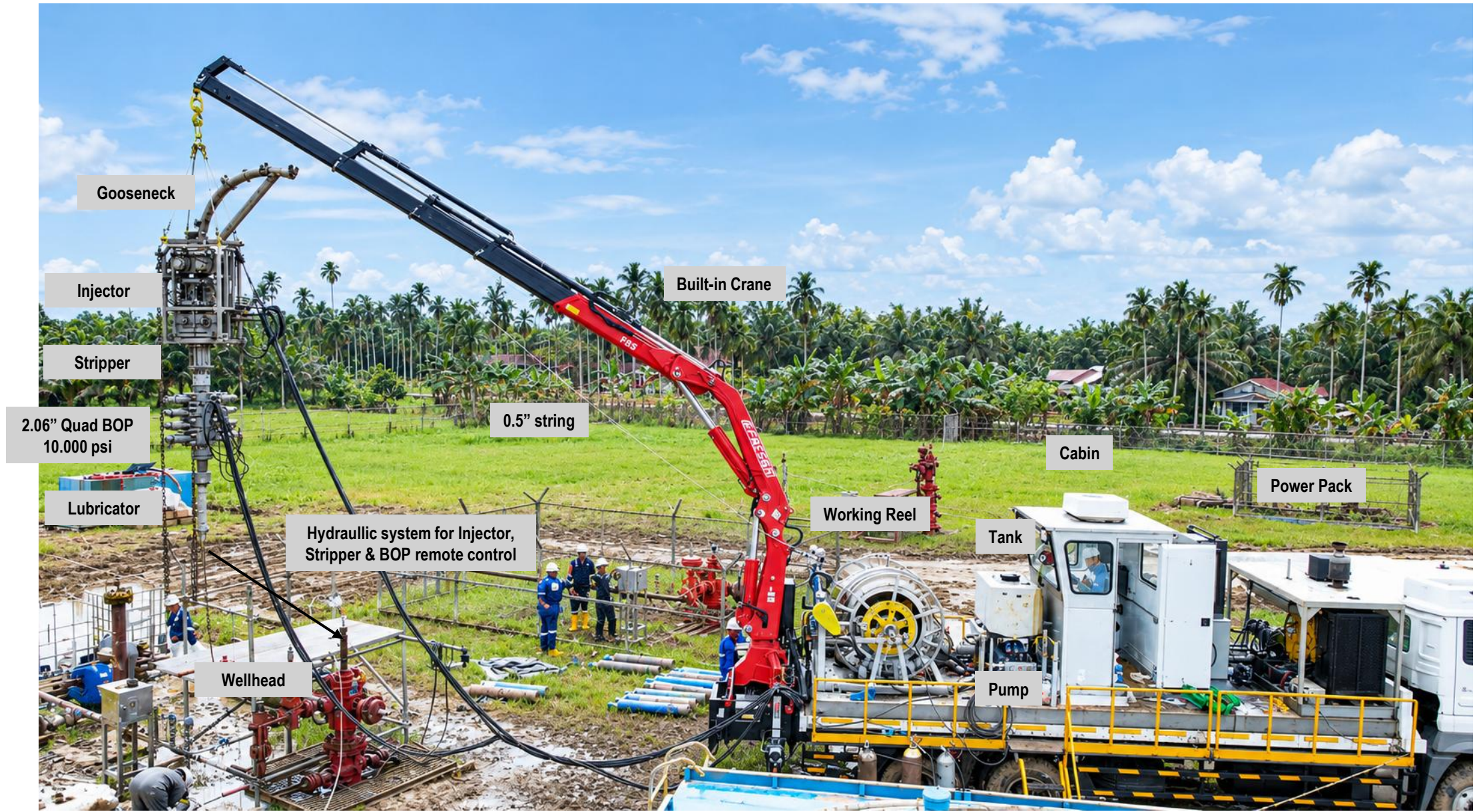
Control
Panel &
Manual indicator

Monitoring Screen with NESR DARA (Data Acquisition & Realtime Analyser)

Pumping Rate, Pressure, WHP, Weight,
Depth, Nitrogen Rate and other required
job Parameters.



LIU – Wellsite Execution



Permanent Chemical Injection



Capstring installation for various purpose chemical injection :

- Foaming agent (liquid loaded gas wells)
- Scale inhibitor
- Corrosion inhibitor
- PPD

Tool / BHA Deployment



Main equipment for tools deployment :

- Pressure chamber monitoring
- Tool installation
- Fishing
- Tools Intervene
- DH Memory gauge

Well Live intervention



- N2 unload
- Low Rate High Conc. Acidizing
- Sand consolidation chemical pumping
- Other stimulation.

Case History - 1



LIU For Well Cleaning and Unloading Services – Prima Energi Bawean, 2026

|| Background

- Idle Oil well due to heavy fluid blockage to the potential and un-intorduced formation.
- Require Formation isolation and unloading
- Our objective is to close the SSD, Unload, and perform Memory logging to gain information of the new formation

|| Result

- SSD successfully closed and isolate unwanted layer, proved with no communication between annulus and tubing
- Well last production parameters reach 400 bopd after unloading services.
- Complete data insight of pressure and temperature gradient, gamma ray, CCL, and Spinner.

|| Execution

- Complete to perform Mechanical SSD closure – N2 unloading, and MPTS logging
- No Issue during the execution in offshore condition.
- Usually done with Slickline, Coil Tubing, and Wireline. Done faster with LIU in one job sequence.



Case History - 2

LIU For Capillary Installation & Chemical Injection – Liquid Loaded Wells, Sanga-Sanga 2021-now

|| Background

- Mature gas field , liquid loading issue
- Initial rate per well = 0.1 -0.2 MMSCFD
- Client practice to overcome liquid loading issue is surfactant pumping

|| Execution

- Injecting surfactant foaming agent to top of surface using capillary string
- Installation using LIU , remove at the top and connect to surface facility (small injection pump)
- 8000 gals of Surfactant for 20 wells



|| Result

- Increase production to 1-1.5 mmSCFD
- Become sustainable and continuous supply for Kalimantan Sanga-Sanga operation

Remove Cap before connect



3.5 EUE
Connection
To Packoff
Assembly



Injector bottom
Connection
(removable)
To sit above packoff
During installation.

Remove & Replace
With hanger after
Capstring
Reach depth

3.5 EUE
towellhead



Case History - 3



LIU For Live Intervention Application – N2 Unload & Simulation , Musi Rawas 2021

|| Background

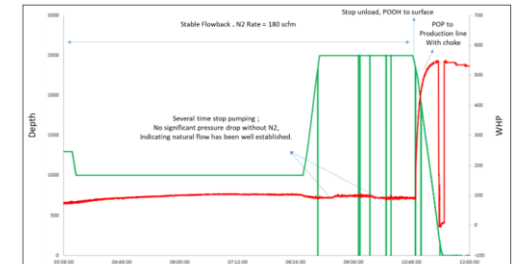
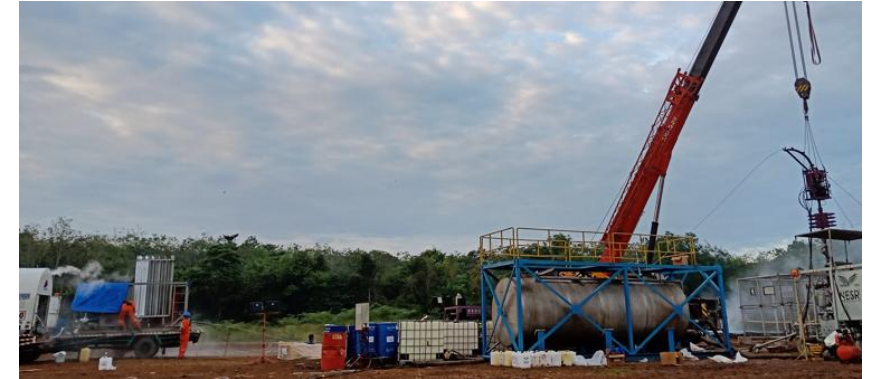
- Tight Damaged formation - Gas
- Acid Bull-heading couldn't be performed due to high pressure
- Lowest rate is required
- Long interval must be exposed
- Required N2 unload before production
- CT is good treatment candidate, but high cost

|| Execution

- 25 bbls HCl for first initiate before bullheading ; rate 0.05-0.08 bpm
- Followed by N2 unload after acidizing
- N2 unload using N2 cryogenic pump and evaporator unit
- N2 rate = 200 scfm
- Total nitrogen 500 gals until stable rate achieved
- Unload time = 5 hours

|| Result

- Successfully stimulate formation without any restriction on rate limitation
- Successfully perform nitrogen unload with small-compact-size equipment configuration
- Production gain to 1,1 MM from initially dead damaged well with Project cost saving 35% from initial plan



Case History - 4



LIU For Capillary Installation & Chemical Injection – Sokoria Geothermal, 2021

|| Background

- New geothermal well with acidic steam
- Require Chemical injection point Downhole
- Our objective is to install the system and pre-test the system with LIU
- High temperature & Harsh Environment
- Depth 1500 m

|| Result

- 3 wells has permanent capstring installation with correct depth and tested successfully
- Capstring installation as their mitigation on having wells with acidic steam production (prevent corrosion problem)

|| Execution

- Install 3 wells in 3 days operation
- No RIH issue
- Usually with slickline , but faster and more reliable with LIU . Injector is one of the improved point.
- Safely Finish installation both downhole and surface
- Injection test prior comissioning





THANK YOU



PT. NPS Energy Indonesia
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Prepared for JDS